

Table 1: Digital friction experiment (7-day, all-hybrid LLM configuration): world-level outcomes and paired effects vs baseline

Panel A. Absolute outcomes across worlds (mean $\pm$ SD over 3 seeds)

World	Attempt rate	Success rate	Help-seek rate	Abandon rate	Neg. share	Helplessness Δ
Baseline (Low Friction)	0.724 $\pm$ 0.015	0.391 $\pm$ 0.019	0.357 $\pm$ 0.010	0.099 $\pm$ 0.006	0.460 $\pm$ 0.022	19.099 $\pm$ 1.672
High Friction + Low Assist	0.399 $\pm$ 0.029	0.044 $\pm$ 0.008	0.218 $\pm$ 0.021	0.122 $\pm$ 0.014	0.889 $\pm$ 0.020	46.784 $\pm$ 1.346
High Friction + High Assist	0.664 $\pm$ 0.057	0.281 $\pm$ 0.025	0.436 $\pm$ 0.041	0.123 $\pm$ 0.020	0.577 $\pm$ 0.008	29.170 $\pm$ 2.314
Low Friction + High Assist	0.821 $\pm$ 0.023	0.573 $\pm$ 0.010	0.480 $\pm$ 0.009	0.042 $\pm$ 0.002	0.301 $\pm$ 0.022	1.811 $\pm$ 1.859

Panel B. Paired effects vs baseline (mean difference [95% CI], n=3 pairs)

Contrast vs baseline	Δ Attempt rate	Δ Neg. share	Δ Helplessness	Δ Trust	Δ Avoidance
High Friction + Low Assist	-0.325 [-0.340, -0.316]	+0.429 [+0.413, +0.439]	+27.686 [+25.850, +29.913]	-26.392 [-28.275, -23.175]	+17.717 [+15.625, +20.200]
High Friction + High Assist	-0.060 [-0.129, -0.009]	+0.118 [+0.100, +0.146]	+10.071 [+7.983, +13.391]	-8.958 [-9.800, -8.000]	+8.192 [+5.825, +9.725]
Low Friction + High Assist	+0.097 [+0.090, +0.109]	-0.159 [-0.177, -0.149]	-17.288 [-18.433, -15.357]	+30.475 [+23.725, +36.475]	-27.458 [-29.175, -26.000]

Notes. Experiment setup from manifest/config: 4 worlds, 3 paired seeds (101/201/301), 10 agents per world, single steady stage, 7 days, 30-minute logical-clock decision interval. All LLM psychology switches are enabled, with `PROTO_LLM_PSYCHOLOGY_MODE=hybrid` and `PROTO_LLM_UNCONTROLLABILITY_MODE=hybrid`. All paired QC rows pass (`qc_pass=1`). Wilcoxon p-values in paired stats are 0.25 for nonzero contrasts (small-sample limit at n=3), so Panel B is primarily descriptive about effect direction and magnitude.